



**FACULTY OF ENGINEERING AND TECHNOLOGY
ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT
ADVANCED DIGITAL DESIGN ENCS3310
COURSE PROJECT**

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Smart Traffic Light Controller with Pedestrian Request : FSM Design and Verification

This project guides you through the process of designing a digital system based on a written specification, converting it into a state diagram, and implementing it in Verilog. You will then verify your design against a reference model using a self-checking testbench.

You are required to design a finite state machine (FSM) that controls traffic lights for a main road, a side road, and a pedestrian crossing. The controller must follow the specification below and be verified against a behavioral reference model provided in this document.

1. System Specification

- Two traffic roads: Main (M) and Side (S).
- One pedestrian button (PB) controlling a crosswalk.
- A vehicle sensor (VS) on the side road.
- Traffic lights for both roads (RED, YELLOW, GREEN).
- Main road is green by default.

Functional Behavior:

1. Default: M-GREEN / S-RED.
2. If VS=1 or PB=1, finish M-GREEN, then M-YELLOW → ALL-RED → S-GREEN (or pedestrian walk).
3. Side-GREEN lasts a fixed time, then S-YELLOW → ALL-RED → M-GREEN.
4. Pedestrian requests during side-green are queued for the next cycle.
5. Timers control minimum durations for green and yellow lights.

Each green state has a time out before checking next state. M at least 60s (may stay green as long as VS=0 and PB=0). S exactly 40s exactly, and P exactly 40s.

2. Interface Definition

```
module traffic_ctrl(  
    input    clk, rst,  
    input    VS, PB,  
    output [2:0] light_main, // {RED,YEL,GRN}  
    output [2:0] light_side, // {RED,YEL,GRN}  
    output    walk    // pedestrian walk signal {RED ,GRN}  
);
```

3. Design Tasks

- Step 1: Read the specification carefully and draw a clear FSM state diagram.
- Step 2: Choose your state encoding (binary, one-hot, or Gray).
- Step 3: Implement your FSM in Verilog with counters for timing.
- Step 4: Use the reference model to verify your RTL design.
- Step 5: Write a self-checking testbench that compares both models.

4. Deliverables

1. FSM state diagram (hand-drawn or digital).
2. Verilog implementation of traffic_ctrl.
3. Testbench and simulation logs.
4. Waveform showing transitions.
5. Short report (≤ 4 pages): state encoding, timing behavior, and results.

7. Grading Policy (100 points)

- FSM Diagram & Specification Translation – 10
- State Encoding and Transition Logic – 10
- RTL Implementation and Reset Handling – 25
- Correctness vs Reference Model – 25
- Testbench Quality – 15
- Report & Waveform Analysis – 15

Deadline:

- The report should be submitted before midnight of Friday 09-01-2026.
- Late submission is penalized at a rate of 10% marks per day until 19-01-2026.